

UNIT-3 PHP [ChatGpt]

2 Marks Questions

1. Give the syntax and example for defining a user-defined function in PHP.

Answer:

```
php                                                                    Copy Edit

function functionName(parameters) {
    // code to be executed
}
```

Example:

```
php                                                                    Copy Edit

function greet($name) {
    echo "Hello, " . $name;
}
greet("Alice"); // Outputs: Hello, Alice
```

2. Differentiate between formal parameters and actual parameters in PHP functions.

Answer:

- **Formal Parameters:** These are the variables defined in the function declaration or definition, which act as placeholders for the values the function expects.
Example: `function sum($a, $b) { }` — here `$a` and `$b` are formal parameters.
- **Actual Parameters (Arguments):** These are the real values or variables passed to the function when it is called.
Example: `sum(5, 10);` — here `5` and `10` are actual parameters.

3. What is function scope in PHP?

Answer:

Function scope refers to the visibility and lifetime of variables declared inside a function. Variables declared inside a function are local to that function and cannot be accessed outside it. Similarly, variables outside the function are not accessible inside the function unless declared as global or passed as parameters.

4. What happens if you call a PHP function with fewer arguments than declared in its definition?

Answer:

- If the function has default parameter values for the missing arguments, PHP uses those default values.
 - If no default values are provided, PHP will generate a **fatal error** for missing arguments.
-

5. Describe the concept of default parameter values in PHP functions.

Answer:

Default parameter values allow you to specify a value for a function parameter in the function declaration. If an argument for that parameter is not provided during the function call, the default value is used.

Example:

```
php Copy Edit

function greet($name = "Guest") {
    echo "Hello, " . $name;
}

greet();           // Outputs: Hello, Guest
greet("Alice");    // Outputs: Hello, Alice
```

6. How do you access global variables within a PHP function? Give example

Answer:

To access global variables inside a function, use the global keyword or the \$GLOBALS array.

Example using global:

```
php Copy Edit

$x = 10;

function test() {
    global $x;
    echo $x; // Outputs: 10
}

test();
```

Example using \$GLOBALS:

```
php Copy Edit

$x = 10;

function test() {
    echo $GLOBALS['x']; // Outputs: 10
}

test();
```

7. What is “pass by value” in the context of PHP function parameters.

Answer:

Pass by value means that a copy of the variable’s value is passed to the function. Changes made to the parameter inside the function do not affect the original variable outside the function.

Example:

```
php Copy Edit

function increment($num) {
    $num++;
}

$a = 5;
increment($a);
echo $a; // Outputs: 5 (unchanged)
```

8. What is "pass by reference" in PHP function parameters?

Answer:

Pass by reference means passing the variable itself, not a copy. Changes inside the function affect the original variable.

This is done by adding an ampersand & before the parameter.

Example:

```
php Copy Edit

function increment(&$num) {
    $num++;
}
$a = 5;
increment($a);
echo $a; // Outputs: 6 (changed)
```

9. What is Default Parameter in PHP functions? Give example

Answer:

A default parameter in PHP is a function parameter that has a predefined value. If no argument is passed for that parameter, the default value is used.

Example:

```
php Copy Edit

function multiply($a, $b = 2) {
    return $a * $b;
}
echo multiply(5); // Outputs: 10 (5*2)
echo multiply(5, 3); // Outputs: 15 (5*3)
```

10. How to check missing parameters in PHP functions? Give example

Answer:

You can check missing parameters by:

- Defining default values for parameters. If parameters are missing, defaults are used.
- Using `func_num_args()` and `func_get_args()` functions to check the number and values of arguments passed dynamically.

Example using `func_num_args()`:

```
php Copy Edit

function test() {
    $num_args = func_num_args();
    if ($num_args < 2) {
        echo "Missing parameters";
    } else {
        echo "All parameters provided";
    }
}

test(1);          // Outputs: Missing parameters
test(1, 2);       // Outputs: All parameters provided
```

Certainly! Here are the questions exactly as you gave, with concise answers suitable for about 4 marks each, covering all possible types/details where applicable.

11. What are anonymous functions? Give examples

Answer:

Anonymous functions, also called closures, are functions without a name, often used as values for variables or as arguments to other functions.

Example:

```
php Copy Edit

$greet = function($name) {
    echo "Hello, " . $name;
};

$greet("Alice"); // Outputs: Hello, Alice
```

12. Differentiate single quoted and double quoted strings in PHP

Answer:

Aspect	Single Quoted Strings	Double Quoted Strings
Variable Parsing	Variables are not parsed (no interpolation)	Variables inside the string are parsed and replaced with their values
Escape Sequences	Only \ and ' are recognized	Supports many escape sequences like \n, \t, \r, \ etc.
Performance	Slightly faster due to no parsing	Slightly slower due to parsing variables and escape sequences

Example:

```
php Copy Edit

$name = "Alice";
echo 'Hello $name'; // Outputs: Hello $name
echo "Hello $name"; // Outputs: Hello Alice
```

13. What is variable interpolation? Give example

Answer:

Variable interpolation is the process where variable values are directly inserted into a string.

Example:

```
php Copy Edit

$name = "Alice";
echo "Hello, $name"; // Outputs: Hello, Alice
```

14. Explain the two ways to interpolate variables into strings?

Answer:

1. **Double-quoted strings:** Variables inside double quotes are automatically parsed and their values inserted.
Example: "Hello, \$name"
2. **Complex (curly) syntax:** Variables can be enclosed in {} to specify boundaries or for complex expressions inside strings.
Example: "Hello, {\$name}!"

15. What are Here Documents in PHP? Give example

Answer:

Here Documents (Heredoc) is a way to define multi-line strings in PHP without using quotes, preserving formatting and allowing variable interpolation.

Syntax:

```
php Copy Edit

$text = <<<EOD
This is a
multi-line string
with variable interpolation: $name
EOD;
echo $text;
```

Example:

```
php Copy Edit

$name = "Alice";
echo <<<EOD
Hello, $name
Welcome to PHP!
EOD;
```

16. Differentiate echo and print statements in PHP

Answer:

Feature	echo	print
Return Value	No return value (void)	Returns 1 (can be used in expressions)
Number of Arguments	Can take multiple parameters	Takes only one argument
Speed	Slightly faster	Slightly slower
Usage	Commonly used for output	Can be used in expressions

Example:

```
php Copy Edit

echo "Hello", " ", "World"; // Valid
print "Hello World";        // Valid
```

17. What is use of print_r() statement? Give example

Answer:

print_r() prints human-readable information about variables, especially arrays and objects.

Example:

```
php Copy Edit

$array = array("a" => 1, "b" => 2);
print_r($array);

/* Output:
Array
(
    [a] => 1
    [b] => 2
)
*/
```

18. What is type specifiers? List the printf type specifiers?

Answer:

- **Type specifiers** are placeholders used in formatted output functions like printf() to define how a value should be formatted.

Common printf type specifiers:

- %d or %i — Integer
- %u — Unsigned integer
- %f — Floating-point number
- %s — String
- %c — Character
- %x — Hexadecimal (lowercase)
- %X — Hexadecimal (uppercase)
- %o — Octal number
- %% — Literal percent sign

19. What is use of var_dump() statement? Give example

Answer:

var_dump() displays structured information about a variable including its type and value. It is useful for debugging.

Example:

php

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```
$var = array(1, 2, 3);  
var_dump($var);
```

```
/* Output:  
array(3) {  
    [0]=> int(1)  
    [1]=> int(2)  
    [2]=> int(3)  
}  
*/
```

20. Explain trim function in PHP along with its types?

Answer:

trim() removes whitespace (or other specified characters) from the beginning and end of a string.

Types of trim functions:

- trim() — removes from both ends
- ltrim() — removes from the left/start only
- rtrim() — removes from the right/end only

Example:

php

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```
$str = " Hello World ";  
echo trim($str); // Outputs: "Hello World"  
echo ltrim($str); // Outputs: "Hello World "  
echo rtrim($str); // Outputs: " Hello World"
```

Here are the questions exactly as you gave, with concise answers for each (about 4 marks), covering all key points and examples:

21. How can you include one PHP file into another PHP file?

Answer:

You can include one PHP file into another using:

- include 'filename.php'; — includes file, raises warning if file not found but script continues.

- `require 'filename.php';` — includes file, raises fatal error if file not found, script stops.
- `include_once 'filename.php';` — includes once, avoids multiple inclusions.
- `require_once 'filename.php';` — requires once, avoids multiple inclusions.

Example:

php

Copy Edit

```
include 'header.php';  
require_once 'config.php';
```

22. Name three commonly used PHP string manipulation functions.

Answer:

- `strlen()` — returns string length
- `str_replace()` — replaces all occurrences of a search string with a replacement
- `substr()` — extracts a part of a string

Other common ones: `strpos()`, `strtolower()`, `strtoupper()`

23. Explain `strcasecmp()` with example?

Answer:

`strcasecmp()` compares two strings **case-insensitively**. Returns 0 if equal, <0 if first is less, >0 if first is greater.

Example:

php

Copy Edit

```
echo strcmp("Hello", "hello"); // Outputs: 0 (equal ignoring case)  
echo strcmp("abc", "Abd");    // Outputs: negative value (abc < Abd)
```

24. Differentiate `strcmp()` and `strcasecmp()` in PHP?

Answer:

Function	Case Sensitivity	Description
<code>strcmp()</code>	Case-sensitive	Compares first n characters of two strings case-sensitively
<code>strncasecmp()</code>	Case-insensitive	Compares first n characters case-insensitively

Example:

```
php                                                                    Copy Edit

strcmp("Hello", "HELLO", 3);    // Not zero, because case-sensitive
strncasecmp("Hello", "HELLO", 3); // Zero, case-insensitive comparison
```

25. Write the syntax of `substr_count()` PHP with example

Answer:

Syntax:

```
php                                                                    Copy Edit

substr_count(string $haystack, string $needle, int $offset = 0, int $length = NULL): int
```

Example:

```
php                                                                    Copy Edit

echo substr_count("Hello world, hello!", "hello"); // Outputs: 1 (case-sensitive)
```

26. Write the syntax of `substr_replace()` PHP with example

Answer:

Syntax:

```
php                                                                    Copy Edit

substr_replace(string $string, string $replacement, int $start, int $length = NULL): string
```

Example:

```
php                                                                    Copy Edit

echo substr_replace("Hello World", "PHP", 6, 5); // Outputs: Hello PHP
```

27. What is the purpose of the substr() function in PHP? Give example

Answer:

substr() returns a part (substring) of a string starting from a specified position and optional length.

Example:

```
php Copy Edit  
  
echo substr("Hello World", 6);    // Outputs: World  
echo substr("Hello World", 0, 5); // Outputs: Hello
```

28. How can you concatenate strings in PHP? Give example

Answer:

Use the dot (.) operator to concatenate strings.

Example:

```
php Copy Edit  
  
$str1 = "Hello";  
$str2 = "World";  
echo $str1 . " " . $str2; // Outputs: Hello World
```

29. List two PHP library functions for handling date and time.

Answer:

- date() — formats a local date and time
- time() — returns current Unix timestamp (seconds since January 1 1970)

Other common functions: strtotime(), mktime()

30. How do you create an instance of a class in PHP? Give example

Answer:

Use the new keyword to create an object.

Example:

php

Copy Edit

```
class Car {  
    public $color;  
}  
$car1 = new Car();  
$car1->color = "red";
```

31. How you access object properties in PHP? Give example

Answer:

Use the arrow operator -> to access properties or methods of an object.

Example:

php

Copy Edit

```
class Person {  
    public $name;  
}  
$p = new Person();  
$p->name = "Alice";  
echo $p->name; // Outputs: Alice
```

32. Define overloading in the context of PHP classes.

Answer:

In PHP, **overloading** refers to the ability to dynamically create properties and methods on objects at runtime using magic methods like `__get()`, `__set()`, `__call()`, and `__isset()`. It is not the same as method overloading in other languages (which is compile-time polymorphism).

Example:

php

Copy Edit

```
class Test {  
    private $data = [];  
    public function __set($name, $value) {  
        $this->data[$name] = $value;  
    }  
    public function __get($name) {  
        return $this->data[$name] ?? null;  
    }  
}  
$obj = new Test();  
$obj->foo = "bar"; // Dynamically setting property  
echo $obj->foo; // Outputs: bar
```

33. What is a constructor in PHP classes? How it is declared?

Answer:

A **constructor** is a special method automatically called when an object is created. It is used to initialize object properties.

Declared using the magic method `__construct()`.

Example:

```
php Copy Edit

class Car {
    public $color;
    public function __construct($color) {
        $this->color = $color;
    }
}

$car = new Car("red");
echo $car->color; // Outputs: red
```

34. What is a destructor in PHP classes? How it is declared?

Answer:

A **destructor** is a special method automatically called when an object is destroyed or script ends. It is used to perform cleanup tasks.

Declared using the magic method `__destruct()`.

Example:

```
php Copy Edit

class Test {
    public function __destruct() {
        echo "Object destroyed";
    }
}

$obj = new Test();
// Destructor called automatically at script end or unset($obj)
```

35. How do you access object properties and methods within a class in PHP? Give example

Answer:

Within the class, use the `$this` keyword with the arrow operator `->` to access properties and methods.

Example:

```
php Copy Edit

class Person {
    public $name;
    public function setName($name) {
        $this->name = $name;    // Access property
    }
    public function greet() {
        echo "Hello, " . $this->name;    // Access property
    }
}

$p = new Person();
$p->setName("Alice");
$p->greet();    // Outputs: Hello, Alice
```

36. What is a trait in PHP, and how does it differ from classes and interfaces?

Answer:

A **trait** is a mechanism for code reuse in PHP that allows you to include methods in multiple classes without inheritance.

Differences:

- **Trait:** Contains method implementations, used to reuse code; can be used in multiple classes.
- **Class:** Defines full objects including properties and methods; supports inheritance.
- **Interface:** Defines method signatures without implementations; classes implement interfaces.

Example:

```
php Copy Edit

trait Logger {
    public function log($msg) {
        echo "Log: $msg";
    }
}

class User {
    use Logger;
}

$user = new User();
$user->log("User created");    // Outputs: Log: User created
```

37. What is introspection in PHP?

Answer:

Introspection in PHP refers to the ability to examine the type or properties of objects or classes at runtime. PHP offers built-in functions and classes like `get_class()`, `get_class_methods()`, `get_object_vars()`, and Reflection classes for introspection.

Example:

```
php Copy Edit

class Test {
    public $var = 1;
    public function foo() {}
}

$obj = new Test();
print_r(get_class_methods($obj)); // Lists methods of the object
```